

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: _____

PRN / Roll No: _____

Division: _____

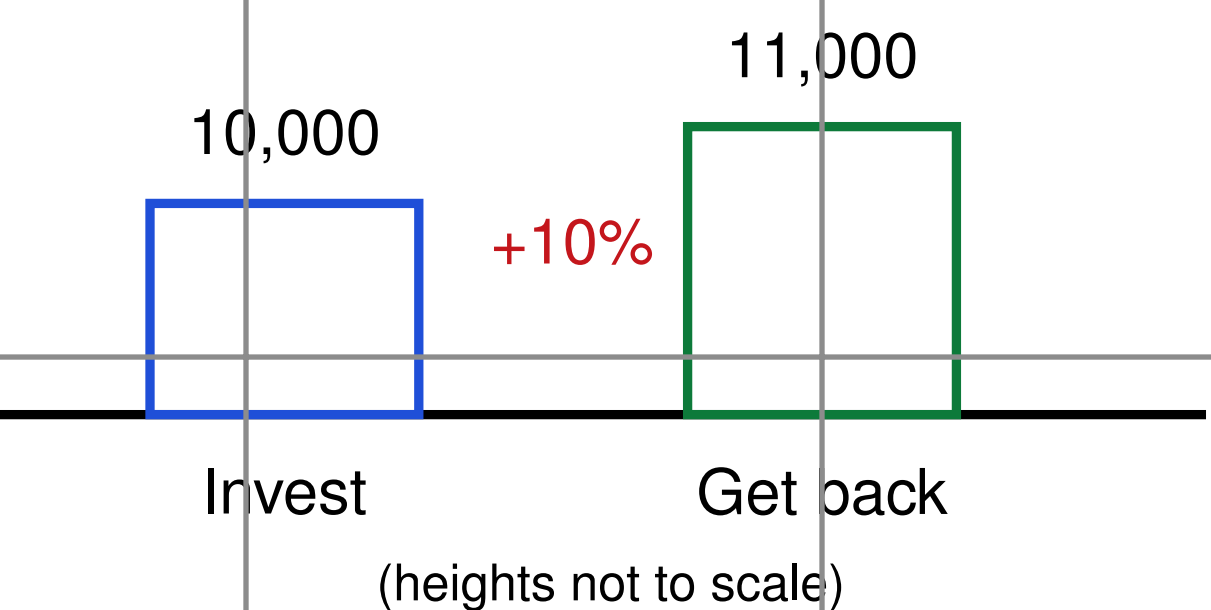
UNIT V: BASICS OF INVESTMENT

1. Investment Return

Gain as a % of the money invested.

$$\text{Return} = \frac{\text{Gain}}{\text{Invested}} \times 100$$

Eg: shares Rs 10,000 → Rs 11,000
Return = $1,000 \div 10,000 = 10\%$



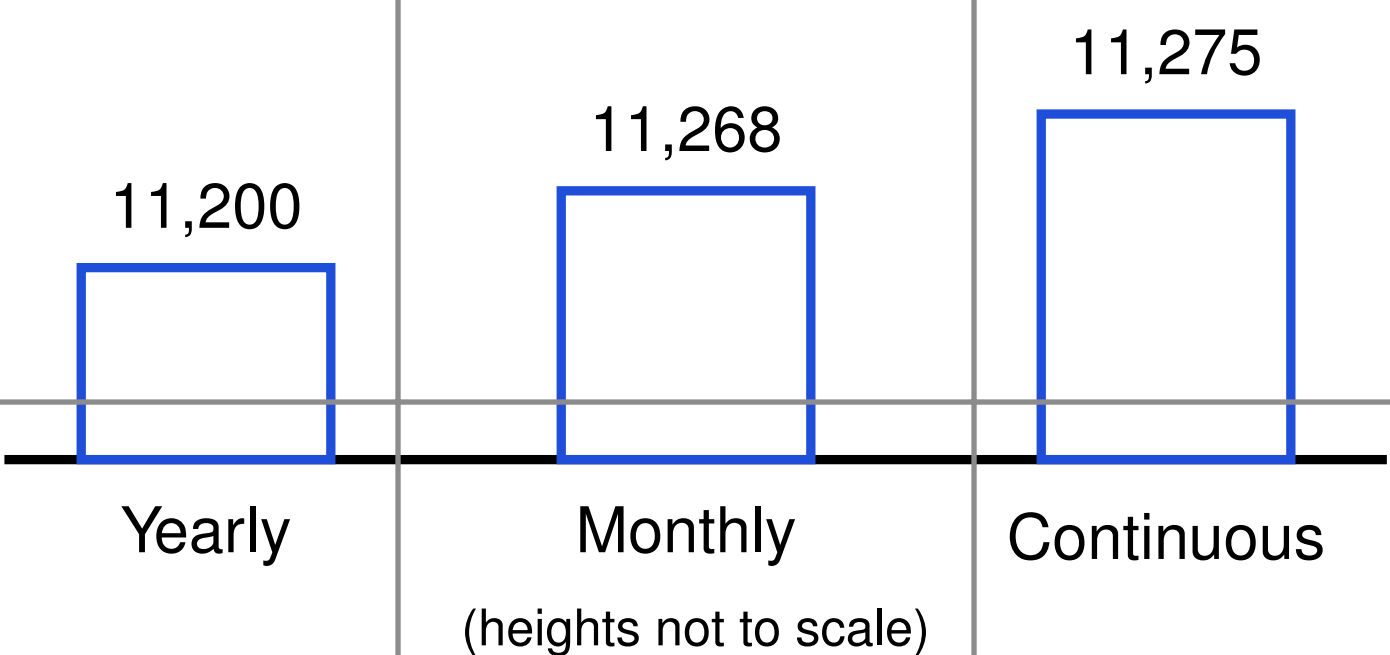
2. Compounding Frequency

More frequent interest = a little more money.

$$A = P \left(1 + \frac{r}{m} \right)^{mt}$$

Continuous: $A = Pe^{rt}$

Eg: bank FD, Rs 10,000 at 12% for 1 year

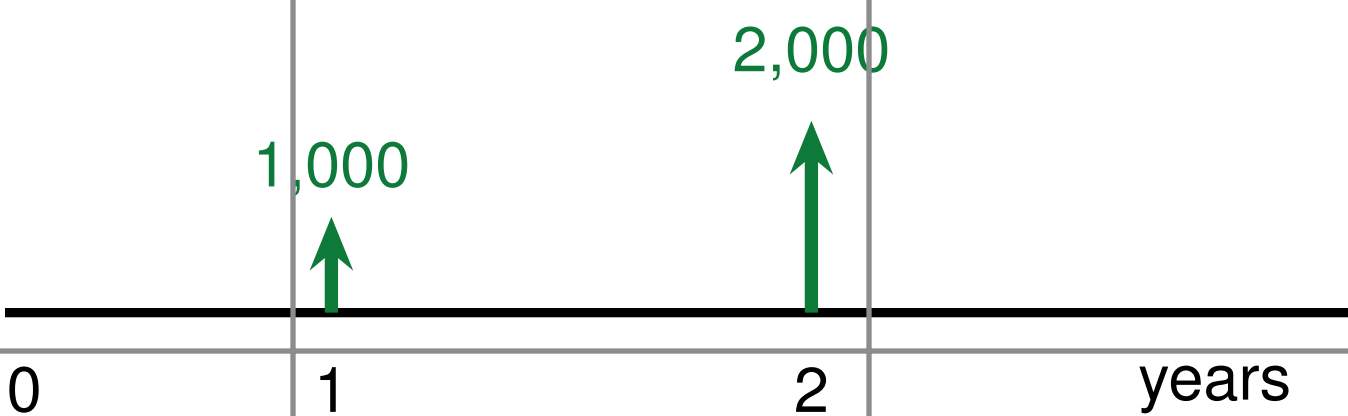


3. Uneven Cash Flows

Unequal receipts: discount each to today.

$$PV = \sum \frac{CF_t}{(1+i)^t}$$

Eg: rent Rs 1,000 then Rs 2,000, $i = 10\%$
 $PV = 909 + 1,653 = \text{Rs } 2,562$

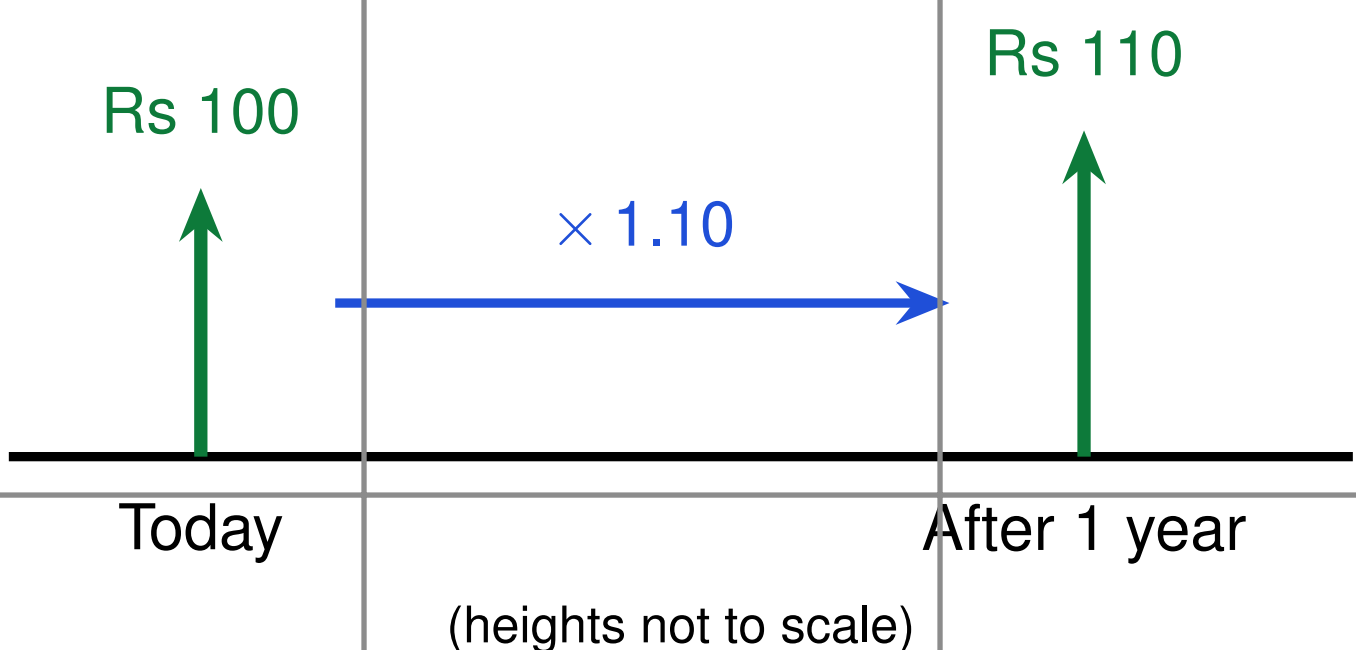


4. Economic Equivalence

Equal value at a given interest rate.

$$F = P(1+i)^n \quad P = \frac{F}{(1+i)^n}$$

Eg: at 10%, Rs 100 now = Rs 110 in 1 yr



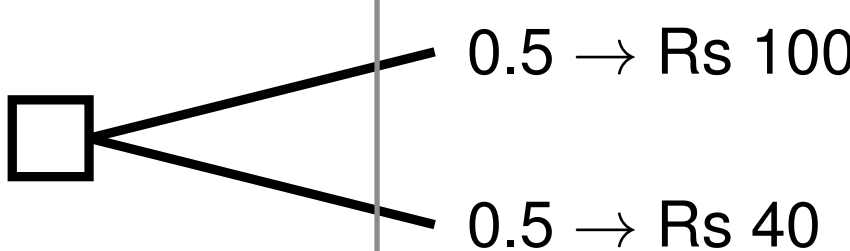
UNIT VI: RISK AND UNCERTAINTY

1. Decision under Risk

Chances known: pick best average (EMV).

$$\text{EMV} = \sum (\text{chance} \times \text{payoff})$$

Eg: $0.5 \times 100 + 0.5 \times 40 = \text{Rs } 70$



2. Decision under Uncertainty

Chances unknown. Use a rule:

- **Maximax**: best of the best → **Stock**
- **Maximin**: best of the worst → **Bond**
- **Min-max regret**: least regret → **Bond**

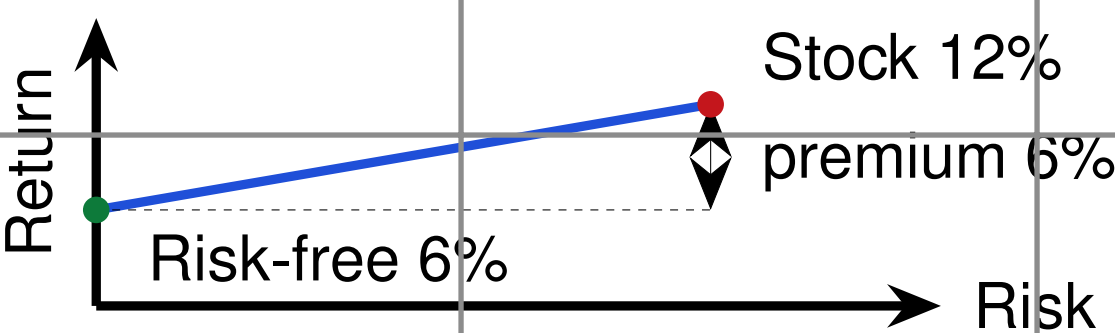
Payoff	Boom	Slump
Stock	80	-20
Bond	40	25

3. Risk Premium

Extra return for taking risk.

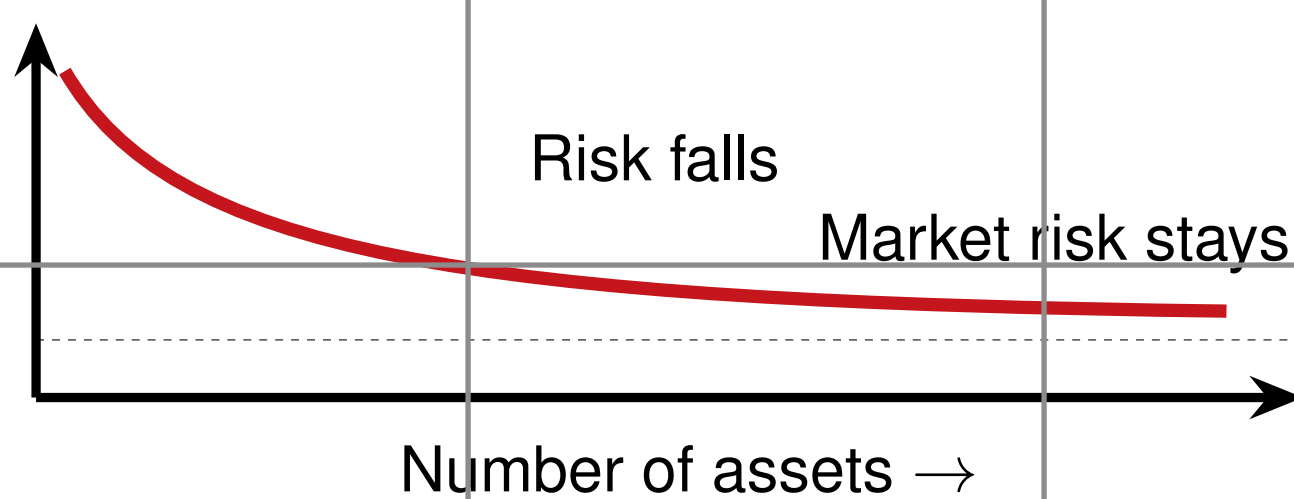
$$\text{Premium} = \text{Expected} - \text{Risk-free rate}$$

Eg: share 12% – FD 6% = 6%



4. Portfolio Diversification

Spread assets: lower risk, not market risk.



5. Life Insurance

Insurer pays if you die within the term.

$$\text{Premium} \approx \text{Sum assured} \times P(\text{death})$$

Eg (1 yr): $\text{Rs } 1,00,000 \times 0.01 = \text{Rs } 1,000$



6. Endowment

Pays on death in term OR survival to end.

$$\text{Endowment} = \text{Term} + \text{Pure endowment}$$

Pure endowment = paid on survival.



Investors demand a higher expected return for taking higher risk